Homeric Echoes in the *Aeneid*: A Machine Learning Analysis Utilizing N-Grams and Logistic Regression

Scott Burke
Georgetown University
Washington, DC, USA
stb62@georgetown.edu

Abstract

The *Aeneid* famously begins with the words *arma virumque cano*. The traditional view in scholarship is that this formulation implies a division between the two halves of Virgil's *Aeneid*: an Odyssean first half, which deals with seafaring and adventuring, with an Iliadic second half depicting the war between the Danaans and the Latins. The notion is present even in ancient commentators on the poem. More recent scholarship in Classics, however challenges the polarized view of the poem, suggesting that the separation of the poems into halves is overly simplistic This paper seeks to use an authorship classification model which utilizes n-gram based logistic regression to see whether the stylistics of the *Aeneid* supports the separation of the poem into an Iliadic and Odyssean half.

1 Introduction

1.1 Classical Scholarship

The assertion that the *Aeneid* can be divided into an Iliadic and Odyssean half is almost as old as the poem itself. Servius the grammarian, a fourth-century commentator on the *Aeneid*, wrote: *primi sex ad imaginem Odyssiae…, hiautem sexadimaginem Iliadosdicti sunt* ("the first six [books of the *Aeneid*] are said to be in the image of the *Odyssey*… the other six are rather said to be in the image of the *Iliad*," Servius on 7.1).[1]

Modern scholars, such as Edan Deckel in Virgil's Homeric Lens and Francis Cairns in Chapter 8 of *Virgil's Augustan Epic*, have provided modern examinations of this claim, seeking to nuance the broad thematic structures with a more complex contours of overlapping thematic investigations. In terms of specific examples of textual reuse, Georg Knaur's seminal *Die Aeneis und Homer* provides a robust comparison of the text itself *Aeneid* with the Homeric poems. This paper seeks to discover how a data-based approach might contribute to this scholarly debate.

1.2 N-Grams in Author Identification

In this paper, I approach the similarities of the *Aeneid* to the Homeric poems as a question of authorship. In other words, we ask the model whether the author of the *Aeneid* more likely to be the author of the *Iliad* or the author of the *Odyssey*. To answer this question, we use POS n-grams with a logistic regression model.

Efstathios Stamatatos' article, "A Survey of Modern Authorship Attribution Methods" compares the effectiveness of a variety of linguistic features to attempt to identify the authors of the Federalist Papers whose authorship is disputed. Among the features they use are both word n-grams and character n-grams. Because we are working across languages, I have opted to use POS n-grams (unigrams, bigrams, and trigrams) to compare stylistic features across the documents. The use of character n-grams would require additional pre-processing of documents and transliteration of the Homeric Greek into characters comparable to the Latin ones of the *Aeneid*. Such a comparison is possible, but beyond the scope of this paper. Jockers and Witten (2010) support the idea that n-grams are one of the most effective tools available to modern computational linguists for authorship identification.

1.3 Some Applications of Author Identification in Classical Texts

Machine learning analyses have been used in other cases to conduct authorship analyses on Classical texts. For example, Milman Parry's groundbreaking work on oral poetics[2] challenged the notion that a "Homer" ever existed, pointing rather to a culture of bards and oral storytellers who constructed

[1] Translation mine.

[2] See the *The making of Homeric verse: the collected papers of Milman Parry*

the Homeric poems over a period of hundreds of years. Pavlopoulos' and Konstantinidou's "Computational authorship analysis of the homeric poems" and Pavlopoulos' "HoLM: Analyzing Linguistic Unexpectedness in Homeric Poetry" both apply methods in computational linguistics to confront the authorship question from a quantitative perspective.

2 Methods

2.1 Corpus

For the *Iliad* and *Odyssey* texts, I used SansDeus' sanitized versions of the Perseus texts (https://github.com/SansDeus/GreekTexts). For the *Aeneid*, I used The Latin Library version of the text from the Classical Language Toolkit (CLTK). The Latin Library texts required post-processing to remove line numbers, headers, and footers, leaving only the Latin text for analysis.

2.2 Stanza and Its Limitations

I tagged each dataset using Stanza, a natural language processing toolkit from Stanford, primarily to obtain POS tags for training a logistic regression model. The tags produced by Stanza are imperfect and vary by language. In this study, we focus on POS tagging accuracy.

The Ancient Greek model trained on the Perseus treebank achieves approximately 92.41% accuracy for UPOS and 85.13% for XPOS. For Latin, the respective accuracies are 91.40% and 78.09%.

For this coarse analysis of similarities between the poems, these accuracies are acceptable, though they introduce noise into the dataset. Ideally, we would use manually annotated versions of all three texts, but such data is not available. Manual annotation was not feasible within the scope of this project.

2.3 Preparing Data for Analysis: Windows and POS Tags

To account for the formulaic structure of Homeric poetry, where scenes often repeat syntactic patterns characteristic of oral poetics, the data was segmented into windows of ten lines, roughly the size of a short speech or scene. The model was trained to classify authorship at the level of these ten-line windows across both Homeric epics and the *Aeneid*.

To enable stylistic comparison across languages, tokens in each segment were replaced with POS tags generated by Stanza. Lemmatization was not used, as it does not provide a reliable cross-linguistic mapping between Latin and Greek.

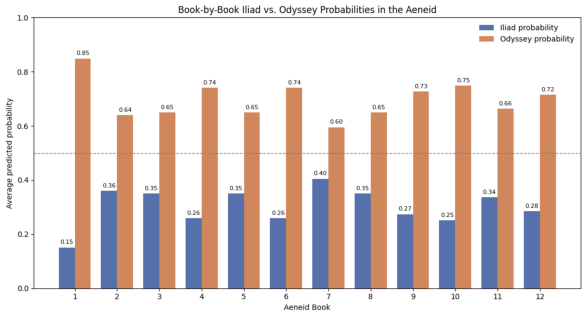

Figure 1: Logistic regression probabilities for the *Aeneid*, i.e. is each book more Iliadic or Odyssean?

2.4 Count Vectorization and Model Training

We used `CountVectorizer` to extract unigram, bigram, and trigram counts of POS tags. Features were standardized using z-scaling to prevent high-frequency tags (such as nouns) from dominating the model.

We then applied Elastic Net regularization to reduce overfitting and mitigate the influence of correlated features.

Finally, we trained a logistic regression model to classify whether a given ten-line segment (represented as POS sequences) was more likely derived from the *Iliad* or the *Odyssey*. This model was then applied to the *Aeneid*, segmented in the same way, to evaluate its stylistic proximity to the Homeric epics. The logistic regression model has an accuracy of 0.864 on unseen data from the *Iliad* and *Odyssey*.

3 Results and Analysis

3.1 Probabilities By Book

The initial findings of the analysis seem to disagree with the classical thesis that the first half of the *Aeneid* is primarily Odyssean and the second half is primarily Iliadic. According to the logistic regression model, the entire poem seems to tend in the Odyssean direction, at least when looking exclusively at POS n-grams. Table 1 contains a breakdown of probabilities by book of how the *Aeneid* compares with the *Iliad* and *Odyssey*. Figure 1 provides a visual representation of that information.

The most notable two books are the extrema, that is, Book 7 of the *Aeneid* being most like the *Iliad* (0.404889 probability) and Book 1 being the

Aeneid Book	*Iliad* Prob.	*Odyssey* Prob.
1	0.151105	0.848895
2	0.360125	0.639875
3	0.349888	0.650112
4	0.258579	0.741421
5	0.350583	0.649417
6	0.259433	0.740567
7	0.404889	0.595111
8	0.350495	0.649505
9	0.273833	0.726167
10	0.251622	0.748378
11	0.335808	0.664192
12	0.284981	0.715019

Table 1: Model probabilities by book for the *Aeneid*.

most *Odyssey*-like (0.848895). Taking a step back and looking at the poem from a thematic perspective, these divisions make sense. Book 1 contains several scenes that have parallels in the *Odyssey*: Aeneas is caught in a storm at sea, shipwrecks in a foreign land, and meets the female rule of the unknown civilization. There are no such scenes in the Iliad. Although there is no immediate connection to be made between these type-scenes and POS n-grams, it is likely that a particular subject of description may in fact affect the syntax with which the scene is depicted.

Book 7 of the *Aeneid* holds a similarly important location in the poem, as it opens the second, supposedly "Iliadic Half". Although the data would appear to indicate that the Iliadic tenor somewhat present in Book 7 does not maintain itself throughout the second half of the book, it at least provides a more Iliadic tenor during a major moment of transition; the poet even evokes the muse as if beginning a second epic poem in full. The book's contents, too, are certainly more Iliadic than Book 1's. The book contains political envoys, the amassing of armies, and the direct involvement of several gods in the action of the poem, which are all more typical of the *Iliad* than they are of the *Odyssey*.

3.2 Mean Scores For Each Half of *Aeneid*

Table 2 depicts the mean score for each half of the *Aeneid*. Both halves of the poem appear to be more like the *Odyssey* than the *Aeneid*. However, looking at how the mean score changes between halves does at least seem to indicate that the second half of the *Aeneid* is slightly more like the *Iliad* in its style than the first half of the *Aeneid*.

Books	Mean *Iliad* Prob.	Mean *Odyssey* Prob.
1–6	0.288286	0.711714
7–12	0.316938	0.683062

Table 2: Mean model probabilities for the first and second halves of the *Aeneid*.

3.3 High-Confidence Example

Let us now examine a case where the model has high confidence. For the following passage, lines 281–290 from Book 1 of the *Aeneid*, the model predicted with 99.95% confidence that the passage was more likely to have come from the *Odyssey*. The passage is included below alongside a translation:

> *consilia in melius referet, mecumque fovebit*
> *Romanos, rerum dominos, gentemque togatam.*
> *sic placitum. veniet lustris labentibus aetas,*
> *cum domus Assaraci Pthiam clarasque Mycenas*
> *servitio premet ac victis dominabitur Argis.*
> *nascetur pulchra Troianus origine Caesar,*
> *imperium Oceano, famam qui terminet astris,*
> *Iulius, a magno demissum nomen Iulo.*
> *hunc tu olim caelo, spoliis Orientis onustum,*
> *accipies secura; vocabitur hic quoque votis.*
>
> ... shall change to better counsels and with me cherish the Romans, lords of the world, and the nation of the toga. Thus is it decreed. There shall come a day, as the sacred seasons glide past, when the house of Assaracus shall bring into bondage Phthia and famed Mycenae, and hold lordship over vanquished Argos. From this noble line shall be born the Trojan Caesar, who shall extend his empire to the ocean, his glory to the stars, a Julius, name descended from great Iulus! Him, in days to come, shall you, anxious no more, welcome to heaven, laden with Eastern spoils; he, too, shall be invoked in vows.[3]

Although there appears to be nothing that makes this passage particularly Odyssean in theme, the style of the passage is very pointed toward the *Odyssey*. We can see in Figure 2 below the features which contribute most to the passage's classification as Odyssean. Notably, punctuation is the primary driver of the classification. The passage above contains 32 PUNCT tags, a notable increase from the mean of 23.15 for all windows in the *Aeneid*. This example seems to suggest that this choppy, heavily punctuated style is typical of the *Odyssey* and is also fairly dominant in the *Aeneid*.

3.4 Low-Confidence Example

Let us now perform a similar examination of a case where the model exhibits low confidence. For the

[3] All translations in this paper come from their respective editions in the Loeb Classical Library.

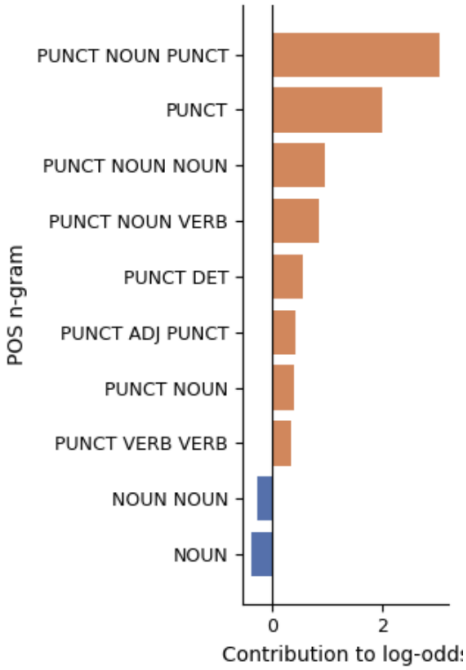

Figure 2: Feature contributions for high-confidence example: (*Aeneid*1.281-290)

following passage, lines 251-260 from Book 12 of the *Aeneid*, the model predicted an almost exact 50/50 split: 0.49984 for the *Iliad* vs 0.50016 for the *Odyssey*. The passage is again included below, along with a translation:

> *arrexere animos Itali, cunctaeque volucres*
> *convertunt clamore fugam (mirabile visu),*
> *aetheraque obscurant pennis hostemque per auras*
> *facta nube premunt, donec vi victus et ipso*
> *pondere defecit praedamque ex unguibus ales*
> *proiecit fluvio, penitusque in nubila fugit.*
> *Tum vero augurium Rutuli clamore salutant*
> *expediuntque manus, primusque Tolumnius augur*
> *"hoc erat, hoc votis" inquit "quod saepe petivi.*
> *accipio agnoscoque deos; me, me duce ferrum...*
>
> The Italians become alert and, wondrous to behold, all the birds clamorously wheel their flight and, darkening the sky with wings, in serried cloud drive their foe through the air till, overborne by the onset and the sheer weight, the bird gave way, dropped the booty from his talons into the stream, and sped deep into the clouds. Then in truth the Rutulians hail the omen with a cheer and spread out their hands. And first of all Tolumnius the augur cries: "This it was, this, that my prayers have often sought! I accept it, I acknowledge the gods. With me, me, at your head, snatch up the sword...

In this case, the model has a hard time distinguishing whether to classify the passage as more Iliadic or Odyssean. Again, punctuation seems to be the primary classifier for the *Odyssey*, while more

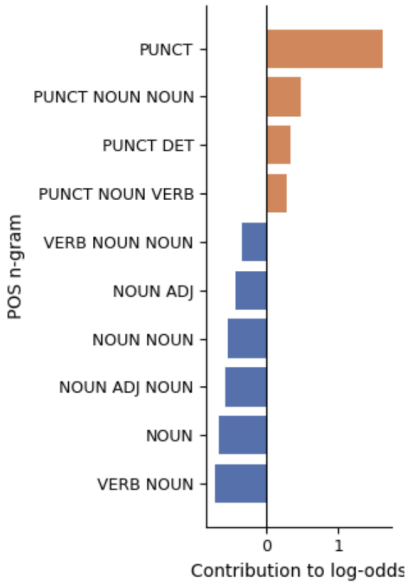

Figure 3: Feature contributions for low-confidence example: (*Aeneid* 12.251-260)

varied POS tags, apparently dominated by nouns (see Figure 3). However, there seems to be fewer features that correlate extremely well with the *Iliad* than with the *Odyssey*. This is aided by the fact that the highest *Iliad* score is a mere .923119 compared with the above passage at 99.5% confidence.

3.5 Examining Punctuation

Table 3 below displays the n-grams most correlated with a passage being classified as more Odyssean.

Of the 20 n-grams which are most indicative of the style of the Odyssey, 17 of them contain punctuation. This is rather strange, considering the data for the Iliad and the Odyssey comes from the same corpus and were tagged using the same Stanza tagger. Additionally, in the original context of the poems, there would have been no punctuation as we know it today. Similarly, because they were originally spoken out loud, there would have been no written punctuation at all.

When we get rid of punctuation from the training corpus, the accuracy plummets by over 20% from 0.863 to 0.629. Perhaps, then the amount of punctuation present in the Odyssey, and how it is used, tells us something about the overall syntactic style of the poem and its similarities to the Aeneid.

4 Conclusion

Ultimately, this paper has used a machine learning analysis to contribute to the debate in Classics scholarship about the structure of Virgil's Aeneid.

Feature	Coefficient	Occurrences
DET PART PART	0.037700	58
PRON PUNCT NOUN	0.039208	65
PUNCT ADJ ADJ	0.046252	122
PUNCT SCONJ	0.054228	1566
DET PART VERB	0.057642	87
PUNCT NOUN	0.068115	3685
PUNCT DET PART	0.070254	398
PUNCT VERB PUNCT	0.070990	158
INTJ ADJ PUNCT	0.073040	47
PUNCT DET	0.076584	524
PUNCT VERB VERB	0.085979	76
PUNCT ADJ VERB	0.088050	168
INTJ NOUN PUNCT	0.124495	57
PUNCT NOUN VERB	0.138801	259
PUNCT ADJ PUNCT	0.156921	140
PUNCT ADJ NOUN	0.158503	382
DET PART	0.167149	516
PUNCT NOUN NOUN	0.272468	350
PUNCT	0.308355	29132
PUNCT NOUN PUNCT	0.358308	243

Table 3: Most Odyssey-like POS n-grams ranked by logistic regression coefficient. Positive coefficients indicate features associated with Odyssey-like classification.

By looking at stylistic features through POS n-grams, this paper demonstrated how, according to a logistic regression classifier, the Aeneid has far more stylistic similarities with the Odyssey than with the Aeneid, although the second half of the Aeneid is more Iliadic than the first. As Figure 1 shows, there is much variation between the individual books of the Aeneid, as well, with the books that start each half (1 and 7) being the most Odyssean and Iliadic, respectively.

Future approaches may use coarser analyses to determine not only stylistic similarities, but also the thematic contours of the poem using a set of categories (exploration, politics, warfare, etc.). There remains much work to be done in this space, and the potential for collaboration between computational linguists and Classicists is nearly unlimited.

Acknowledgments

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